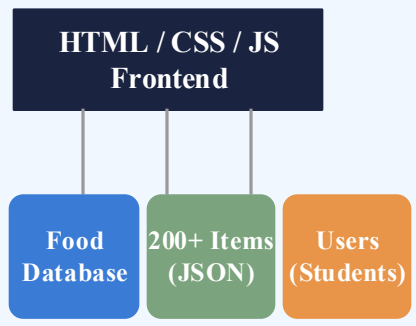
Essential Nutrition For Healthy Diet

PROJECT BASED LEARNING FROM FWD

Abstract

The Essential Nutrition For Healthy Diet is a smart web-based Nutrition Calculator featuring 200 food items across Vegetables, Fruits, Dairy, and Indian Foods. Built with HTML, CSS, and JavaScript, it offers real-time search, category filtering, animated UI, dynamic nutrition calculation, and daily goal progress bars. The app features a floating animated logo, fade-in results, and visual progress tracking for a 2000 kcal daily diet.

System Architecture



— CO-WISE IMPLEMENTATION —

CO1 HTML Document Structure & CSS Basics

- ✓ Responsive container layout with border-radius & box-shadow
- ✓ CSS Animations: floating logo (@keyframes float), fadeIn results
- ✓ Typography: Georgia serif, 46px h1 heading, line-height 1.6
- ✓ Color scheme: linear-gradient(135deg) background and sections
- ✓ Semantic HTML with charset UTF-8, viewport meta, proper title

CO2 CSS Pseudo-classes, Flexbox & Forms

- ✓ Flexbox food-row with flex:2 select and flex:1 quantity input
- ✓ :hover & :focus pseudo-class selectors with outline & box-shadow
- ✓ Category pill filter with .active class toggling via JavaScript
- ✓ Form inputs: select, number type with min=0.5, max=1000, step=0.5
- ✓ 4 animated progress bars tracking daily nutrition goal percentages

CO3 JavaScript Basics, Objects & Loops

- ✓ 200-item foods{} object: name, serving, calories, protein, carbs, fat
- ✓ Array .filter() + .slice(0,8) for real-time food search dropdown
- ✓ forEach() loop generates HTML result cards for each food item
- ✓ Functions: addRow(), searchFoods(), filterCat(), calc()
- ✓ 60 Vegetables, 50 Fruits, 22 Dairy, 68 Indian Foods — 4 optgroups

CO4 DOM Manipulation & Dynamic Updates

- ✓ Dynamic food rows via createElement('div') and appendChild()
- ✓ Live search dropdown toggled via classList.add/remove('show')
- ✓ Nutrition totals written to 8 DOM elements via element.innerText
- ✓ Progress bar widths set dynamically: element.style.width = pct + '%'
- ✓ Category filter hides/shows <option> elements via style.display

CO5 Exception Handling & Input Validation

- ✓ try-catch block wraps the entire calc() nutrition function
- ✓ Throws Error('Please select a food item') if select value is empty
- ✓ Throws Error('Please enter a valid quantity') for zero/missing input
- ✓ catch(e) displays user-friendly alert(e.message) on any error
- ✓ HTML required attribute + min=0.5 / max=1000 attributes on inputs

Features Interactive Features

- ✓ Smart search bar: live 8-result dropdown on every keystroke typed
- ✓ Category pills: All Foods / Vegetables / Fruits / Dairy / Indian
- ✓ Daily goals tracker: Calories/2000, Protein/50g, Carbs/300g, Fat/70g
- ✓ Multi-item meal builder with + Add More and ✕ remove row buttons
- ✓ Animated floating logo (@keyframes float) and fade-in results

Conclusion & Future Scope

This project demonstrates core web development concepts — responsive layout, CSS animations, JavaScript objects & loops, DOM manipulation, and exception handling — through a real-world nutrition tool. Future enhancements include AI-driven meal recommendations, user profiles, calorie tracking history, barcode food scanning, and backend integration with Node.js / Express.js for personalized dietary analysis.